

# Ahsanul Hoque

Founder and CEO, Livora | Software Engineer (Full-Stack, AI and Embedded Systems)

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## SUMMARY

Computer Science undergraduate (University of Chittagong, expected 2026) and Founder/CEO of Livora, a full-stack AI-powered healthcare platform that won 2nd place and an incubation package at the ITBI Student Startup Pitch Fest 2026 (CUET). Ships production-grade full-stack and applied-AI systems across healthcare, e-commerce, embedded/IoT, and data engineering. Comfortable across the stack: React/TypeScript frontends, Node.js/Express APIs, SQL data modeling, LLM/RAG integration, and embedded systems (ESP32, LoRa).

## EXPERIENCE

**Founder and CEO - Livora (Self-employed)** | May 2026 - Present

*Chattogram, Bangladesh (Hybrid)*

- Lead product, engineering, and business strategy for Livora, an AI-powered healthcare platform serving patients, doctors, lab technicians, and hospital administrators
- Won 2nd place at the ITBI Student Startup Pitch Fest 2026 (CUET), securing 6 months of free incubation space, mentoring, and startup-ecosystem support worth BDT 100,000
- Conducted an AI training workshop on the platform for practicing surgeons in Chittagong

## SKILLS

**Languages:** Python, C++, TypeScript, JavaScript, SQL

**Web and Backend:** React, Node.js, Express, Prisma, Sequelize, REST APIs, Socket.IO

**AI / LLM:** LLM Integration (Google Gemini), RAG (Retrieval-Augmented Generation), Prompt Injection Defense, Langfuse (LLM Observability), OCR (Tesseract), Speech-to-Text (Deepgram)

**Data and Infrastructure:** PostgreSQL, MySQL, Redis, Docker, HL7 / FHIR / DICOM (healthcare interoperability), Cloudinary, Pino, Sentry

**Embedded and IoT:** ESP32, STM32, Arduino, FreeRTOS, LoRaWAN, Raspberry Pi, ThingSpeak

**Other:** Data Structures and Algorithms, Competitive Programming, Blockchain Fundamentals, Project Management

## PROJECTS

**Livora - AI-Powered Healthcare Platform** | Oct 2025 - Present

*React, TypeScript, Node.js, PostgreSQL, LLM/RAG, Socket.IO*

- Building a full-stack clinical platform covering appointment booking, voice-driven prescriptions, lab order/result management with payments, DICOM/FHIR and HL7 ingestion, and multi-disciplinary team (MDT) video consultations
- Implemented a persistent AI memory system and a RAG-based health assistant with medicine safety checks, risk stratification, and prompt-injection defenses; instrumented with Langfuse and Sentry

**Vision-Based Autonomous Navigation for a Quadrotor in GPS-Denied Environments** | Aug 2026 - Present

*Python, PyBullet, NumPy, A\*/RRT\*, PID Control, IEEE LaTeX*

- Designed a full perception-planning-control pipeline for GPS-denied quadrotor navigation using only onboard depth-camera vision, simulated in PyBullet
- Benchmarked A\* against RRT\* across 60 trials (3 obstacle densities x 2 planners x 10 trials); A\* won on success rate and path efficiency (1.03x optimal vs 1.06-1.11x) with about 30x lower planning latency (36ms vs approx. 1.1-1.2s)
- Wrote up the full study as a 5-page IEEE conference paper with experimental results and figures generated from the sweep data

**Bengali Medical Dialogue Generation - Nascenia AI Hackathon** | Aug 2026

*Python, PyTorch, LoRA, NLP*

- Fine-tuned an LLM under 3B parameters (LoRA on Qwen3-1.7B) to generate doctor-style responses in Bengali to patient questions for a Kaggle competition hosted by Nascenia
- Evaluated TF-IDF retrieval, retrieval+generation hybrids, semantic retrieval, and model-soup ensembling; documented what worked and why

**Scraper Pipeline - AI-Powered Web Scraping and Extraction System** | Jul 2026 - Present

*Python, Playwright, Anthropic/OpenAI/Gemini/Groq/DeepSeek, pytest*

- Built a 4-stage pipeline (discover, collect, extract, clean) with a cost-aware multi-tier LLM cascade and cross-provider fallback chain that only escalates to a costlier model when deterministic validation of the cheaper tier's output fails
- Implemented Cloudflare/WAF bypass via headless-browser fallback, crash-safe atomic writes, resume-by-default batch processing, and a 92-test no-network suite covering every pure-logic component

**Second Brain - Autonomous AI Agent for Productivity and Automation** | May 2026 - Present

*Claude Code, Python, Telegram Bot API, Git, Cron*

- Built an autonomous AI agent that runs study, startup, and daily-task workflows from a git-versioned Obsidian vault, with a Telegram bot as a mobile interface for logging progress, marking tasks done, and queuing background jobs
- Orchestrated 4 scheduled cron jobs (morning briefing, logging nudge, nightly re-planning, overnight task runner) atop a compiled knowledge-caching layer that processes source material once and serves every later query from cache

### **ResCris - Long-Range Emergency and Crisis Management Network | 2025**

*ESP32, LoRa (SX1278), GPS, Embedded Systems*

- Built an internet-independent emergency communication system using LoRaWAN for disaster response, with real-time SOS reporting, GPS-based tracking, and a multi-gateway architecture
- Presented at Chittagong Science Carnival 4.0 with a teammate

### **Aurora Gadgets - Full-Stack E-Commerce Platform | Apr 2026 - May 2026**

*React, TypeScript, Express, Prisma, MySQL, Docker*

- Built a complete storefront and admin system covering products, categories, brands, cart, and checkout with category-based product attributes
- Containerized the application with Docker for reproducible deployment

### **Agri Supply Chain Transparency Platform | 2025**

*Blockchain*

- Designed a blockchain-based platform to make farm-to-consumer pricing transparent and verifiable, aimed at closing the margin gap between farmer income and consumer prices
- Runner-up, Hackathon Segment, Chittagong Science Carnival (hosted by CUSS)

### **Warehouse Management and Billing System | Freelance**

*Node.js, MySQL*

- Built an inventory, billing, and sales-analytics system for a retail client

### **Sales Data Warehouse and BI Reporting | Aug 2025 - Sep 2025**

*SQL, Python, Jupyter Notebook*

- Designed and built a modern SQL data warehouse from scratch, consolidating sales data via ETL and dimensional modeling
- Built interactive BI dashboards with Python (Pandas, Matplotlib, Seaborn) for trend and segmentation analysis

### **AgroEdge AI - Precision Agriculture Platform | Jan 2026 - Apr 2026**

*Python, Raspberry Pi, ThingSpeak*

- Built an AI-first precision-agriculture stack: data pipeline, irrigation model training, edge inference, and a Raspberry Pi dashboard integrated with ThingSpeak

### **WaifOS - Multitasking OS for ESP32 | Mar 2025 - May 2025**

*C++, Arduino, FreeRTOS*

- Built a lightweight multitasking operating system for ESP32 featuring an app menu, display rendering, and input handling
- Implemented WiFi connectivity and concurrent task scheduling on constrained embedded hardware

## **EDUCATION**

**University of Chittagong - B.Sc. in Computer Science and Engineering | 2022 - 2026 (Expected)**

*CGPA: 3.50/4.00 (through 6th semester)*

**Govt. City College, Chattogram - Higher Secondary Certificate, Science (GPA 5.00) | 2019 - 2022**

**Chittagong Collegiate School - Secondary School Certificate, Science (GPA 5.00) | 2013 - 2019**

## **AWARDS AND ACHIEVEMENTS**

- 2nd Place, ITBI Student Startup Pitch Fest 2026, CUET - for Livora (incl. incubation support)
- 1st Place, Robo Soccer, EEE Fest 2025, University of Chittagong
- Runner-up, Hackathon Segment, Chittagong Science Carnival (hosted by CUSS)
- 1st Runner-up, Robo Soccer, Bangabandhu Innovation Fair 2024, Premier University
- Finalist, Bangabandhu Innovation Grant (BIG), 2023
- Champion, ETE DecaHz - SDG-focused idea competition (algae-based circular economy model)
- 2nd Runner-up, Intra-Department Programming Contest, 22nd Batch

## **LEADERSHIP AND ACTIVITIES**

- President, N-Cube (Science Club), Chittagong Collegiate School
- Joint Secretary, Debating Society, Chittagong Collegiate School